

Forensic Data Quality Assessment and System Redesign for Aadhaar Enrolment Data Governance

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Abstract

Analysis of the UIDAI Aadhaar enrolment and update dataset reveals a clear separation between operational effectiveness and data governance quality. Aggregate enrolment counts exhibit a stable right-skewed, long-tailed log-normal distribution, suggesting coordinated, campaign-driven execution. However, persistent spelling inconsistencies, referential integrity failures, temporal gaps, and administrative hierarchy conflicts indicate systemic weaknesses in data collection and storage design. This report conducts a forensic data quality assessment to identify root causes of these failures and proposes a governance-first database and system redesign to enforce correctness at data entry rather than through downstream cleaning. The redesign also includes a centralized–distributed system architecture, where regional servers enforce uniform schemas and validation at write-time, and periodically synchronize with a central authoritative database using logged, retry-safe aggregation, ensuring consistency without disrupting operational throughput.

Problem Statement and Approach

Problem Statement

The Aadhaar enrolment and update datasets display a persistent gap between aggregate-level coherence and record-level reliability. Although enrolment volumes follow a stable right-skewed, long-tailed log-normal distribution—indicating effective, campaign-driven operational execution—the underlying data contains systematic inconsistencies, including non-normalized administrative names, invalid and conflicting pincodes, missing dates and months, and violations of state–district–pincode hierarchies. These issues occur uniformly across states, datasets, and time periods, demonstrating structural flaws in data collection and aggregation rather than isolated data-entry errors. As a result, the dataset’s usefulness for fine-grained policy analysis, longitudinal monitoring, and regional comparison is significantly compromised despite apparently reliable aggregates.

Approach

This report adopts a forensic, system-design-centric approach rather than a data-cleansing-centric one. Data quality failures in both raw and processed datasets are treated as evidence of missing validation, master data management, and referential integrity controls within the existing system. Statistical behaviour and error distributions are analysed to distinguish operational execution from governance failure. Based on these findings, a governance-first database redesign is proposed, introducing standardized schemas, master reference tables for administrative entities, enforced foreign-key constraints, and write-time validation. The

proposed architecture aims to preserve proven operational processes while preventing the introduction of structural inconsistencies at source.

Datasets Used

1. [Aadhar Enrollment dataset](#)
2. [Aadhar Biometric Updates Dataset](#)
3. [Aadhar Demographic Updates Dataset](#)
4. [All India Pincode Directory](#)

Methodology

The methodology consists of parallel analysis of cleaned and raw datasets to distinguish data-entry artefacts from structural system failures. Each analysis step is designed to map observed inconsistencies to missing governance controls rather than to correct the data.

The entire process of analysis was done in two categories:

1. [Analysis of cleaned data](#)
2. [Analysis of raw data](#)

Analysis of cleaned data

This analysis involved the following steps:

1. [Data Cleansing](#)
2. [Preprocessing and Inspection](#)
3. [Visualization of records - Month-wise and State-wise](#)
4. [Z-Score and Ratio Analysis](#)
5. [Daily records - Histogram analysis](#)
6. [Pincode Validity and Coverage Analysis](#)

Data Cleansing:

Process:

Text data was normalized so that different spellings of the same entity, such as State, were merged into a single value. Data pertaining to invalid State records were deleted and, wherever the State name was replaced by the district or area name, the corresponding State name was applied.

Inference:

The State and District columns are not present in normalized form. They have various different spellings, etc. pertaining to the same entity.

This implies that the Master Data used to govern this dataset is either incomplete or non-existent.

Preprocessing and Inspection:

Preprocessing:

- Datasets were loaded into [Huggingface](#) from the [APIs provided](#) (because the 3 zip files provided in the portal had incomplete data, only almost 3/4ths of the complete API data)
- Date format was corrected (from String to Date)
- New Year-Month columns was added to identify the month
- State data was cleaned
- Jan-2026 records were removed

Inspection:

- Columns and record count of the pre-processed file was found
- Unique States and Months were found

Inference:

- The given data in the form of Zip files is incomplete or generated through wrong querying.
- Date format is handled incorrectly

Visualization of Records - Month-wise and State-wise totals:

Process:

Group the data by Year-Month and State. Sum all records of different age groups and plot the sum by Year-Month for each State.

Plot this on a log scale with all states combined into one graph.

Inference:

Total records are very erratic with sudden 200-300% increases and drops, which means a lot of data is missing.

Z-Score and Ratio Analysis:

Process:

- Aggregate demographic counts by State×Month and Month (overall).
- Z-score normalize per-state time series and the overall monthly series.
- Merge on month.
- Compute relative standardized deviation(z_{state}/z_{total})
- Plot state-wise temporal ratios.

Inference:

Most states have a ratio of around -1 to 1 for the most part. This means, the data inconsistencies in the individual states are all due to the same causes as the entire dataset and not one-off mistakes in a few individual states.

Daily Records - Histogram Analysis:

Process:

- Find the list of daily enrollment / update counts in total and for each state and plot those on a logarithmic scale
- Find the number of unique dates in each month and state during which records exist

Inference:

Frequencies of daily enrollment / update counts follow a right-skewed, long-tailed log-normal distribution. This indicates that enrolments occur in concentrated bursts, consistent with organised drives or policy-led campaigns rather than steady continuous activity.

However, in some months, data is present for every day of the month including mandatory government holidays.

Whenever data exists, it exists for all states, which implies that data is not recorded properly on some days during aggregation from centres throughout the country.

Again, the issue is not at the individual state level, but at the final data compilation level.

Pincode validity and Coverage analysis:

Process:

Set of unique pincodes from the given Aadhar datasets and the All India Pincode Registry dataset were compared.

Inference:

There are around 700-1000 pincodes present in the All India Pincode Registry but absent in the Aadhar datasets. This implies that either many areas do not have the facilities for Aadhar enrollment / updates or, more likely, there is an issue with data collection.

More importantly in the converse case, there are around 700-1000 pincodes present in the Aadhar datasets but absent in the All India Pincode Registry. This implies that there is no single standardized master database for pincode data.

Analysis of raw data

This analysis involved the following steps:

1. [Detection of One Pincode -> Multiple States / Districts mappings](#)
2. [State names audit](#)

Detection of One Pincode -> Multiple States / Districts mappings:

Process:

For each pincode, record all instances of different district or state names.

Inference:

Some pincodes are mapped to multiple districts or states, not just because of spelling mistakes.

State names audit:

Process:

Record all unique state names including different spellings. Map them to actual state.

Inference:

30% of the states have multiple spellings in the dataset

It is clear that a unified structure is not being followed while collecting or aggregating data.

Code

<https://github.com/an-42-an/uidai1>

Data Analysis and Visualization

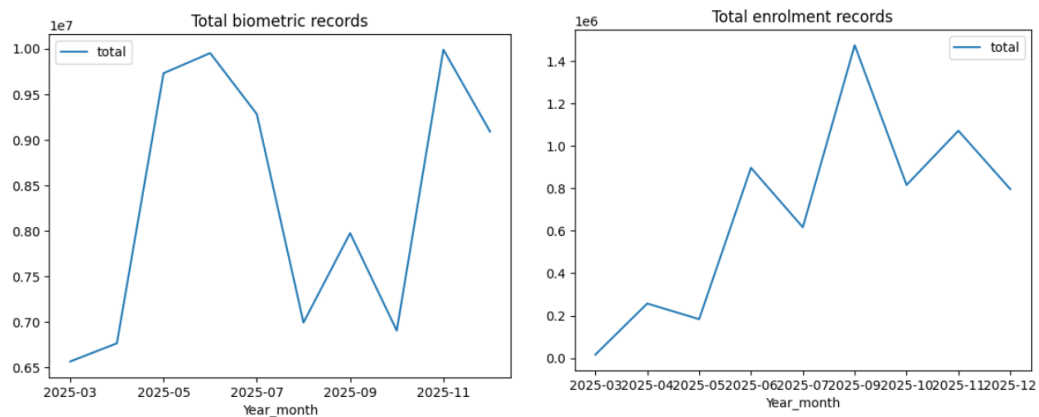
Given below is a consolidation of all inferences and their supporting visual evidence

1. Given ZIP files data is incomplete - doesn't match the given official API data

	ZIP Files Record Count	API Data Record Count
Demographic	20,71,700	23,75,882
Biometric	18,61,108	50,00,000
Enrollment	10,06,029	12,08,727

2. Erratic total records across months - incomplete data

For more states's visualizations, visit [here](#)



3. Data inconsistencies in individual states are all due to the same causes as the entire dataset and not one-off mistakes in a few individual states.

For more states's visualizations, visit [here](#)

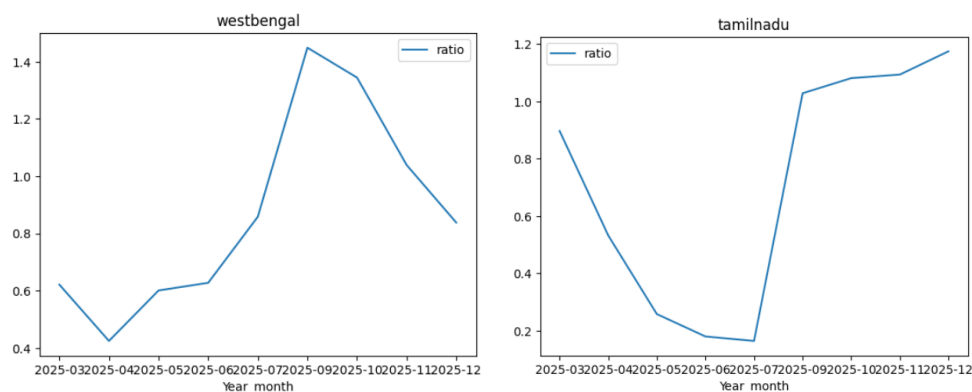


Figure: Z-score ratio between each state and total count across states

Most states have a ratio of around -1 to 1 for the most part. This means, the data inconsistencies in the individual states are all due to the same causes as the entire

dataset and not one-off mistakes in a few individual states.

4. Frequencies of daily enrollment / update counts follow a right-skewed, long-tailed log-normal distribution. This indicates that enrolments occur in concentrated bursts, consistent with organised drives or policy-led campaigns rather than steady continuous activity.

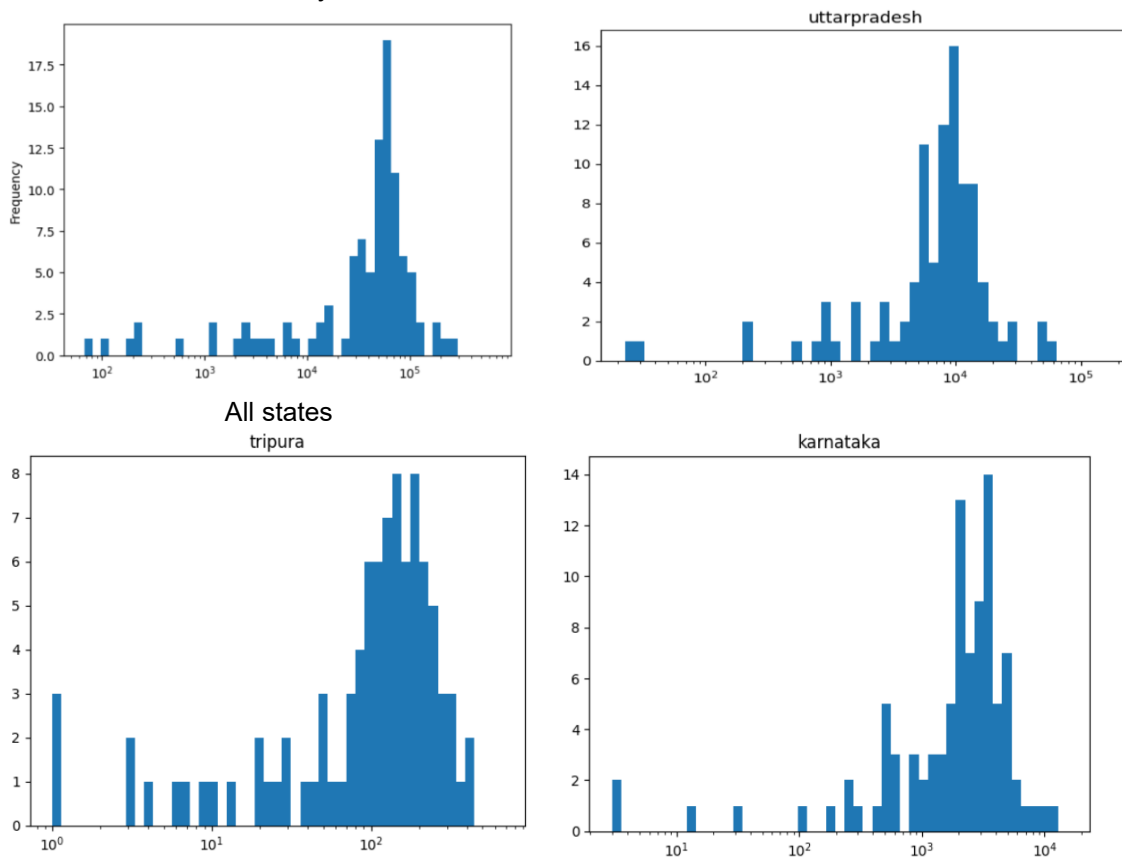


Figure: frequencies of daily enrollment
For more states's visualizations, visit [here](#)

5. Data absent for $\frac{1}{3}$ - $\frac{2}{3}$ of the year but sometimes present even on mandatory government holidays

Table: No. of unique dates per month during which records exist

Enrollment	Biometric	Demographic
Date	Date	Date
Year_month	Year_month	Year_month
2025-03 9	2025-03 1	2025-03 1
2025-04 1	2025-04 1	2025-04 1
2025-05 1	2025-05 13	2025-05 1
2025-06 12	2025-06 30	2025-06 1
2025-07 1	2025-07 31	2025-07 1
2025-09 30	2025-08 31	2025-09 20
2025-10 18	2025-09 30	2025-10 18
2025-11 20	2025-10 31	2025-11 30
2025-12 11	2025-11 30	2025-12 30
	2025-12 29	
Total: 103 days	Total: 227 days	Total: 103 days

6. When data exists, it exists for all states. This implies that data is not recorded properly on some days during aggregation from centres throughout the country.

Year_month	2025-03	2025-04	2025-05	2025-06	2025-07	2025-09	2025-10	2025-11	2025-12
State									
andamanandnicobarislands	1.0	NaN	NaN	NaN	NaN	20.0	18.0	30.0	30.0
andhrapradesh	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
arunachalpradesh	1.0	NaN	NaN	NaN	NaN	20.0	18.0	30.0	30.0
assam	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
bihar	1.0	NaN	NaN	NaN	1.0	20.0	18.0	30.0	30.0
chandigarh	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
chhattisgarh	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
dadranagarhavelidamandiu	1.0	1.0	NaN	NaN	NaN	20.0	18.0	30.0	30.0
delhi	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
goa	1.0	NaN	NaN	NaN	NaN	20.0	18.0	30.0	30.0
gujarat	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
haryana	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
himachalpradesh	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
jammuandkashmir	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
jharkhand	1.0	1.0	1.0	NaN	NaN	20.0	18.0	30.0	30.0
karnataka	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
kerala	1.0	NaN	NaN	NaN	NaN	20.0	18.0	30.0	30.0
ladakh	1.0	NaN	NaN	NaN	NaN	20.0	18.0	29.0	30.0
lakshadweep	NaN	NaN	NaN	NaN	NaN	18.0	17.0	30.0	30.0
madhyapradesh	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
maharashtra	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
manipur	1.0	NaN	NaN	NaN	NaN	20.0	18.0	30.0	30.0
meghalaya	1.0	NaN	NaN	NaN	NaN	20.0	18.0	30.0	30.0
mizoram	1.0	1.0	1.0	1.0	NaN	20.0	18.0	30.0	30.0
nagaland	1.0	NaN	NaN	NaN	NaN	20.0	18.0	30.0	30.0
odisha	1.0	NaN	NaN	NaN	1.0	20.0	18.0	30.0	30.0
puducherry	1.0	NaN	1.0	NaN	1.0	20.0	18.0	30.0	30.0
punjab	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
rajasthan	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
sikkim	1.0	NaN	NaN	NaN	NaN	19.0	18.0	30.0	30.0
tamilnadu	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
telangana	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
tripura	1.0	NaN	NaN	NaN	NaN	20.0	18.0	30.0	30.0
uttarakhand	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
uttarpradesh	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0
westbengal	1.0	1.0	1.0	1.0	1.0	20.0	18.0	30.0	30.0

Table: Unique dates in enrollment dataset

For more raw data, visit [here](#)

7. Many pincodes present in the Aadhar dataset are missing in the All India Pincode Registry and vice versa.

For more raw data, visit [here](#)

Dataset	Present in dataset but not in All India Pincode Registry	Present in All India Pincode Registry but not in dataset
Demographic	995	775
Biometric	1028	746
Enrollment	829	919

8. Some pincodes are mapped to multiple districts or states, not just because of spelling mistakes.

Example:

Look at the following few pincodes mapped to multiple states or districts

Pincode	State	District
160014	Chandigarh, Punjab	Chandigarh, Rupnagar, S.A.S Nagar(Mohali), SAS Nagar (Mohali)
151302	Punjab	Bathinda, Mansa

For more raw data, visit [here](#)

	Pincodes with >1 districts	Pincodes with >1 states
Enrollment	6715	1325
Biometric	8532	1755
Demographic	7786	1529

9. 30% of the states have multiple spellings in the dataset

For more raw data, visit [here](#)

	States with multiple spellings	Invalid States
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Biometric	10	12
Demographic	9	13
Enrollment	10	1

Sources of Data Inconsistencies

Collection of Location Data:

State and district names are entered as free text, which allows the same place to appear under multiple spellings or formats. Since there is no fixed list to choose from, consistency depends on the person entering the data, leading to repeated naming variations.

Lack of Validation During Data Entry:

Pincodes and dates are accepted without being checked against any official reference. This allows invalid, missing, or incorrectly formatted values to be stored in the dataset and reused in later records.

Absence of Administrative Hierarchy Checks

There is no system-level check to ensure that a district actually belongs to the selected state or that a pincode matches the stated location. This results in conflicting geographic relationships within the same record.

Aggregation Masks Underlying Issues

When records are aggregated by month or state, totals appear smooth and consistent. This hides record-level errors and gives a false impression of overall data quality, allowing problems to persist unnoticed.

Proposed Solution:

Governance-First Database with Distributed System Architecture

By introducing a Database Schema for immediate inconsistency check during data entry, it is possible to decrease discrepancies within the aggregate data. Validation is enforced at write-time, ensuring structural correctness before data is persisted.

Relational database systems (RDMBS) must be implemented with referential keys spanning across all tables to maintain data integrity. The implementation of this is as follows:

- Introduce a governance-first relational database design that enforces correctness at data entry rather than relying on post-hoc data cleaning.

- Replace free-text administrative fields with master reference tables for states, districts, and pincodes, linked via enforced foreign-key constraints.
- Standardize schemas across enrolment, biometric, and demographic datasets to enable consistent aggregation without additional cleaning.
- Enforce write-time validation for administrative hierarchy and temporal completeness to prevent structurally invalid records from being stored.
- Preserve existing operational workflows while ensuring record-level integrity and long-term analytical reliability.

Standardized Schema

We propose a new database schema with common columns for all datasets, with separate tables for:

- States - State ID, State Name
- Districts - District ID, District Name, State ID
- Pincodes - Pincode, District ID
- Biometric - Date, Pincode, Age 5-17, Age 17+
- Enrollments - Date, Pincode, Age 0-5, Age 5-17, Age 17+
- Demographic - Date, Pincode, Age 5-17, Age 17+

The age columns may be stored in this aggregated way or stored as the raw integer age and later be aggregated.

This ensures consistency across enrolment, biometric and demographic records, reducing spelling errors, duplicate entries and mismatched administrative data.

Referential Integrity:

All location fields are linked to master tables for states, districts, and pincodes. This prevents invalid combinations and maintains hierarchy across datasets.

Aggregation-Friendly Design:

By storing all datasets with common columns, monthly and state-wise aggregation can be done reliably without additional cleaning, while preserving record-level correctness.

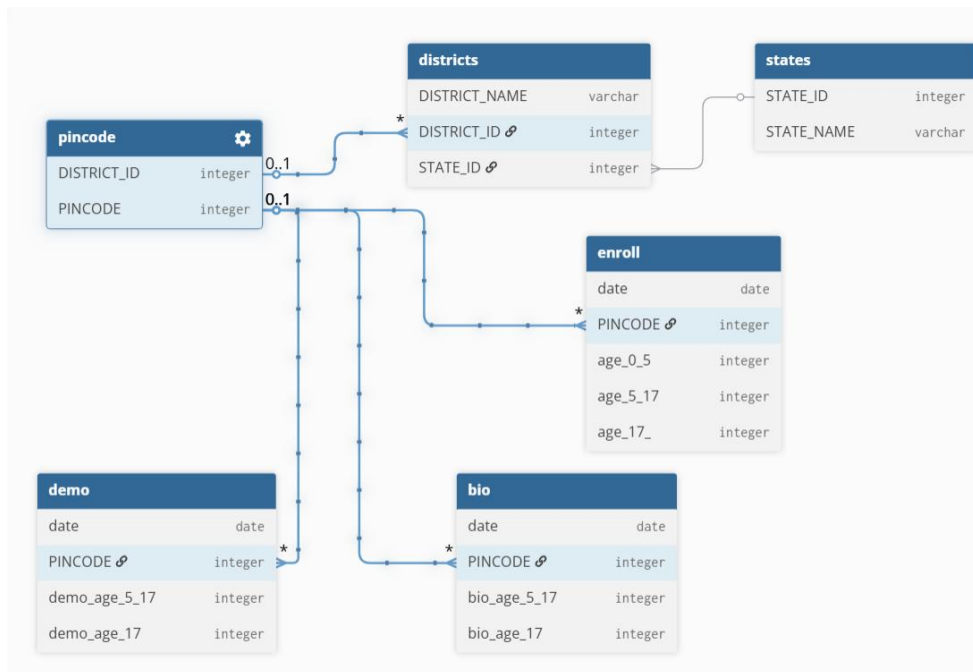


Figure: Proposed Database Schema

Distributed System Architecture

The data-governance system is designed as a distributed architecture to ensure scalable, consistent, and reliable processing of Aadhaar enrolment and updation data across the country. Data integrity is enforced at ingestion rather than corrected during downstream analysis.

Ingestion, Validation, and Master Data Enforcement

Enrolment centres submit records through secure APIs to stateless ingestion services. These services validate schema compliance, mandatory fields, data formats, and enforce authoritative state–district–pincode mappings through centralised master data services. Hierarchical and temporal validity checks are performed at write-time to prevent invalid administrative combinations.

Buffered Processing and Storage

A message queue or stream-processing layer buffers incoming records, absorbing burst traffic during large-scale enrolment campaigns without data loss. Validated records are stored in a normalised transactional database with enforced foreign-key constraints, which serves as the authoritative source of truth.

Analytics, Aggregation, and Recovery

Data is periodically replicated in an append-only manner to an analytics warehouse for reporting and aggregation, ensuring analytical workloads do not affect ingestion performance. Aggregation and backup tasks are logged with execution status, and any missed or failed runs are automatically retried during subsequent processing windows to prevent gaps in time-based statistics.

Auditability and Governance

Immutable audit logs track record corrections, reprocessing events, and administrative boundary changes over time, enabling traceability, reproducibility, and long-term data governance.

Summary

- Aadhaar enrolment and update data shows strong, coherent aggregate behaviour consistent with campaign-driven operations, but widespread record-level inconsistencies across geography, time, and administrative hierarchies.
- These inconsistencies are systemic, occurring uniformly across states and datasets, and originate from the absence of enforced master data, validation, and hierarchy checks rather than isolated data-entry errors.
- Data aggregation and downstream cleaning conceal underlying structural failures, giving a misleading impression of data reliability.
- A governance-first relational database redesign, enforcing standardized schemas, master reference tables, and write-time validation, is necessary to preserve operational effectiveness while ensuring record-level correctness.
- The proposed solution includes a centralized–distributed ingestion architecture where regional systems validate data at source and synchronize with a central authoritative database, ensuring scalability without compromising governance.

Conclusion

The Aadhaar dataset demonstrates effective enrolment execution but inadequate data system design. Improving data quality requires architectural enforcement of governance, not downstream correction. The proposed redesign preserves operational strengths while restoring analytical reliability.